

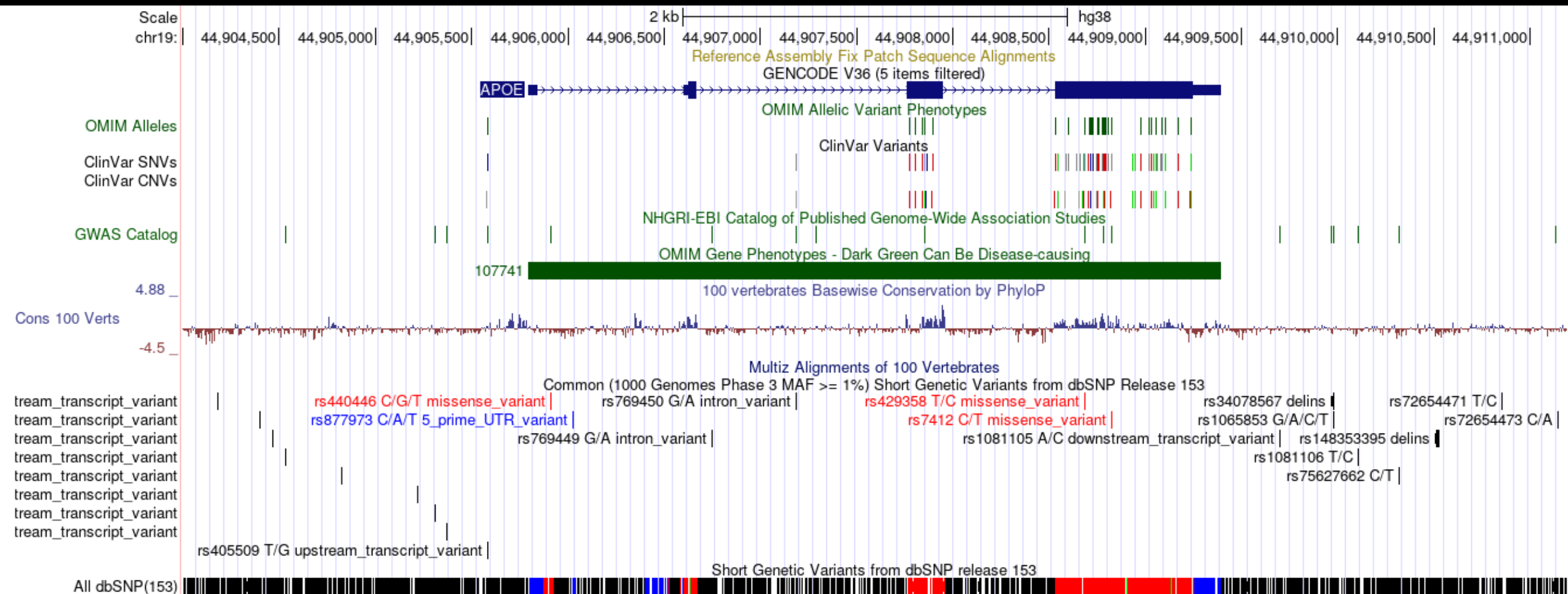
Genome Browser 1

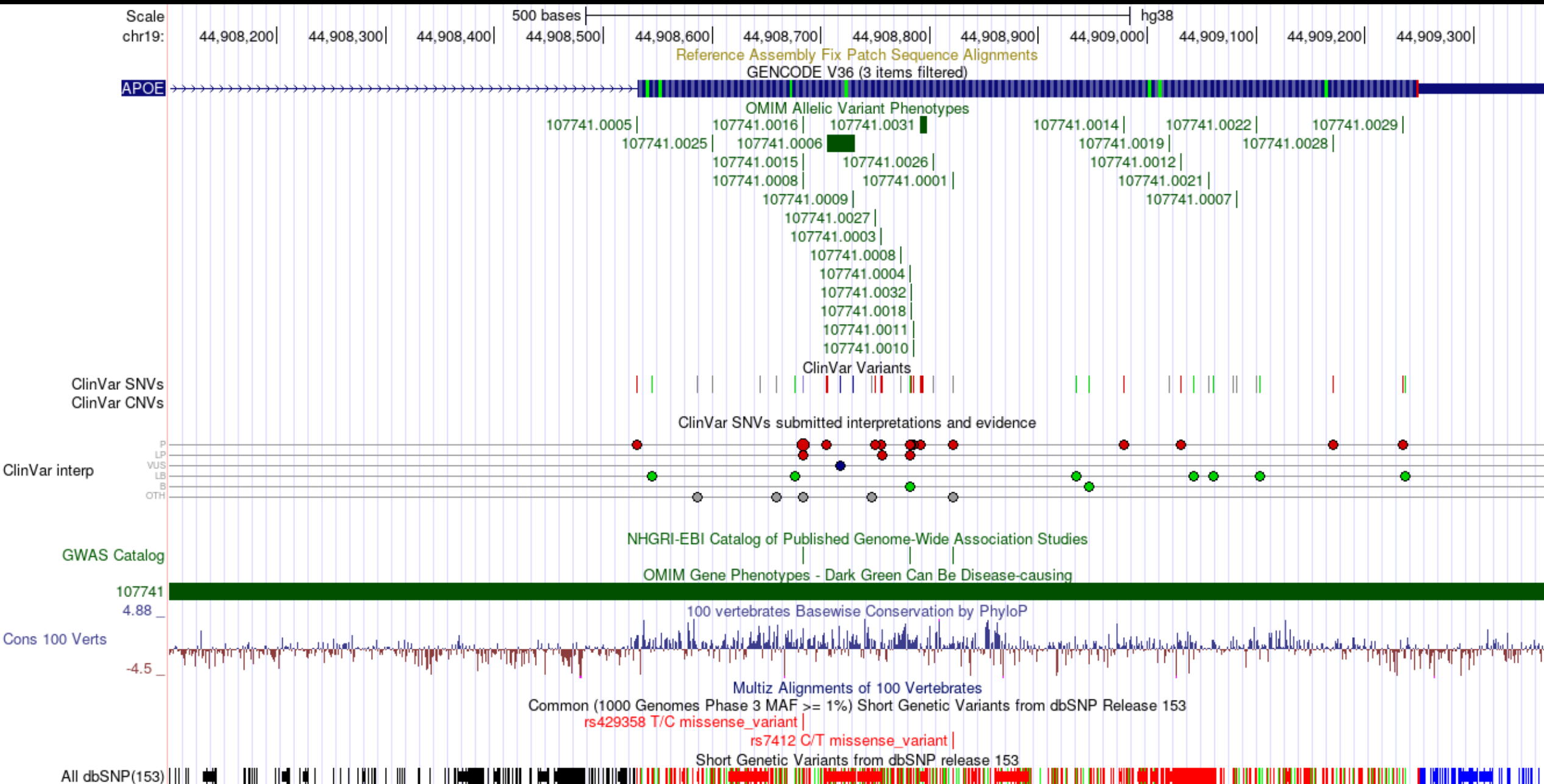
Tools and Tables

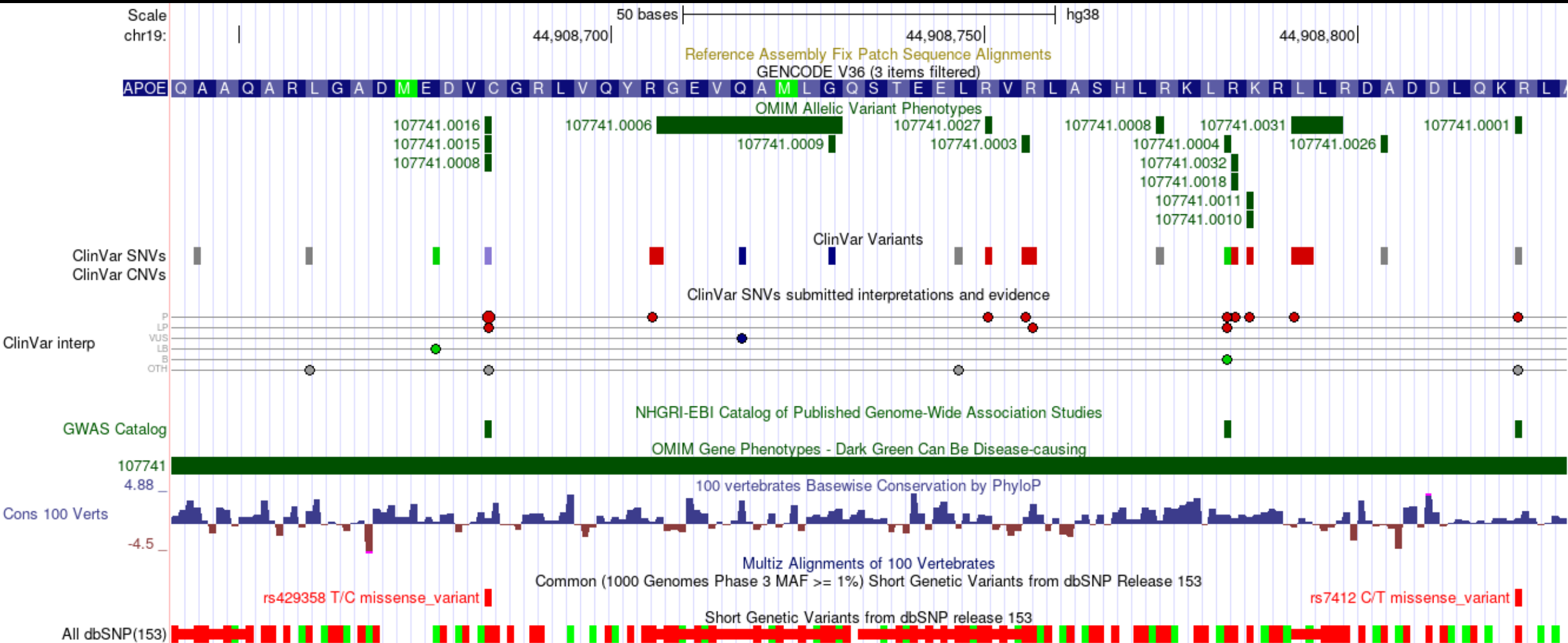
HUGEN 2073

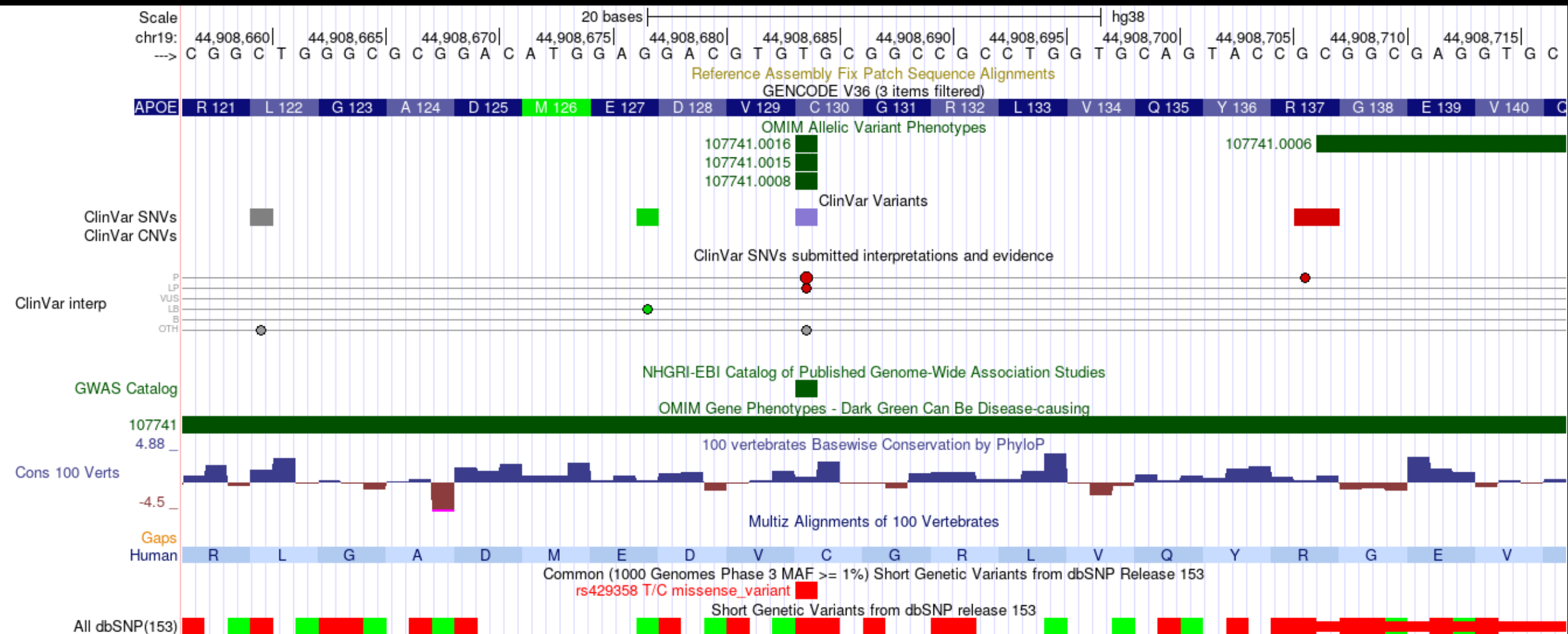
Genomic Data Visualization & Integration

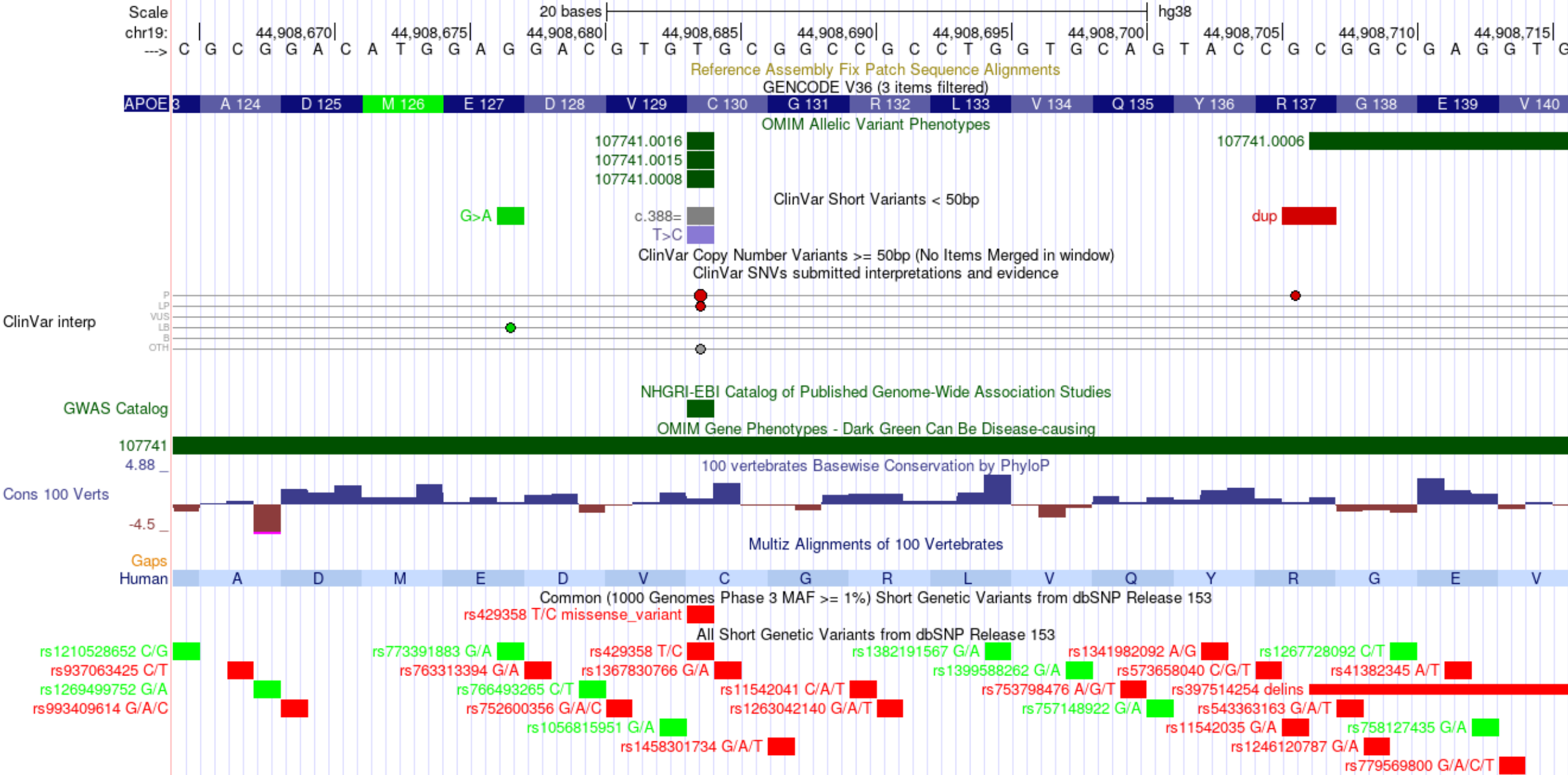
Slides adapted from Ryan L. Minster, PhD, MSIS











University of California at Santa Cruz (UCSC) Genome Browser

- An on-line, and downloadable, **genome browser**
- An interactive website offering access to **genome sequence data** from a variety of vertebrate and invertebrate species and major model organisms, integrated with a large collection of **aligned annotations**.
- An open-source, web-based tool suite built on top of a **MySQL database** for rapid visualization, examination, and querying of the data at many levels.
- The **Genome Browser Database**, browsing tools, downloadable data files, and documentation can all be found on the UCSC Genome Bioinformatics website.

Citations

- Kent *et al.* **The human genome browser at UCSC.** *Genome Res*12, 996 (2002) <https://doi.org/10.1101/gr.229102>
- Kuhn *et al.* **The UCSC Genome Browser Database: update 2009.** *Nucleic Acids Res*37, D755 (2009) <https://doi.org/10.1093/nar/gkn875>
- Fujita *et al.* **The UCSC Genome Browser database: update 2011.** *Nucleic Acids Res*39, D876 (2011) <https://doi.org/10.1093/nar/gkq963>
- Lee *et al.* **UCSC Genome Browser enters 20th year.** *Nucleic Acids Res*48, D756 (2020). <https://doi.org/10.1093/nar/gkz1012>

UCSC Genome Browser Home

genome.ucsc.edu/index.html

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Tools

Genome Browser - Interactively visualize genomic data

BLAT - Rapidly align sequences to the genome

In-Silico PCR - Rapidly align PCR primer pairs to the genome

Table Browser - Download and filter data from the Genome Browser

LiftOver - Convert genome coordinates between assemblies

REST API - Returns data requested in JSON format

Variant Annotation Integrator - Annotate genomic variants

More tools...

News

Mar. 08, 2023 - **TOGA Gene Prediction tracks for 41 assemblies**

Feb. 24, 2023 - **New GENCODE gene tracks: V43 (hg19/hg38) - VM32 (mm39)**

Feb. 13, 2023 - **New recombination rate tracks for hg38**

Jan. 30, 2023 - **Addition of GRCh38 patch 14 sequences to hg38**

Dec. 20, 2022 - **Multiz Alignment & Conservation (470 mammals) for hg38**

Dec. 13, 2022 - **New GENCODE gene tracks: V41 (hg19/hg38) - VM30 (mm39)**

More news...

Subscribe

Sharing data

Click the images to explore the latest **Public Sessions** on our main site.

Sessions - Configure, save, and share short links to Genome Browser views

Track Hubs - A powerful tool for displaying your own or viewing other data annotations

Custom Tracks - A simple way to display some custom annotations

Learning

Training - Visit our training page for videos, manuals, and tutorials on the Genome Browser

Education - We offer teaching modules using the Genome Browser aimed at the undergraduate classroom

Workshops - If you would like to request a virtual or in-person workshop, please **contact us**

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?

Features

- Visual Interface ("**Genome Browser**" on the landing page)
- Underlying Data Tables ("**Table Browser**")
- Two types of data integrators
 - Site Based ("**Variant Annotation Integrator**")
 - Range Based ("**Data Integrator**")
- Desktop version [**"Genome Browser in a Box (GBiB)"**]
- Genome coordinate conversion ("**LiftOver**")
- Ability to add custom tracks ("**Track Hubs**")

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Visual Interface

- Choose organism and assembly
- Enter a search term (optional):
 - gene symbol
 - variant ID
 - range chrN:0-0

The screenshot shows the UCSC Genome Browser Gateway website. The browser tabs include 'UCSC Genome Browser Gateway' and 'Genome Browser FAQ'. The address bar shows 'genome.ucsc.edu/cgi-bin/hgGateway'. The website header features the University of California Santa Cruz Genomics Institute logo and the 'Genome Browser Gateway' title. A navigation bar includes links for Genomes, Genome Browser, Tools, Mirrors, Downloads, My Data, Projects, Help, and About Us.

The main content area is divided into two columns. The left column, titled 'Browse/Select Species', contains 'POPULAR SPECIES' (Human, Mouse, Rat, Zebrafish, Fruitfly, Worm, Yeast) and 'REPRESENTED SPECIES' (a phylogenetic tree with a color-coded bar on the left). The right column, titled 'Find Position', includes a 'Human Assembly' dropdown menu set to 'Dec. 2013 (GRCh38/hg38)', a 'Position/Search Term' input field, and a 'GO' button. Below this, it displays 'Human Genome Browser - hg38 assembly' with a 'view sequences' button. The assembly details section lists: 'UCSC Genome Browser assembly ID: hg38', 'Sequencing/Assembly provider ID: Genome Reference Consortium Human GRCh38.p12 (GCA_000001405.27)', 'Assembly date: Dec. 2013 initial release; Dec. 2017 patch release 12', 'Assembly accession: GCA_000001405.27', 'NCBI Genome ID: 51 (Homo sapiens (human))', 'NCBI Assembly ID: 5800238 (GRCh38.p12, GCA_000001405.27)', and 'BioProject ID: PRJNA31257'. A 'Search the assembly:' section provides instructions for searching by position, gene name, or track type. A 'Download sequence and annotation data:' section lists options: 'Using rsync (recommended)', 'Using FTP', 'Using HTTP', 'Data use conditions and restrictions', and 'Acknowledgments'. The bottom section is titled 'Assembly Details'.

Search Results

Human hg38 LDLR UCSC Genes

Genome Browser FAQ

genome.ucsc.edu/cgi-bin/hgTracks?hgsid=1064495105_sFNkAdpAAoXKGnMmJMO60eGrFe3J&org=Human&db=hg38&position=LDLR&pix=1339

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Your search resulted in multiple matches. Please select a position:

Gencode Genes

[LDLR \(ENST00000558518.6\) at chr19:11089463-11133820](#) - Homo sapiens low density lipoprotein receptor (LDLR), transcript variant 1, mRNA. (from RefSeq NM_000527)
[NR1H2 \(ENST00000652203.1\) at chr19:50358606-50383337](#) - Orphan receptor. Binds preferentially to double-stranded oligonucleotide direct repeats having the consensus half-si
[VLDLR \(ENST00000382099.2\) at chr9:2622100-2654480](#) - Homo sapiens very low density lipoprotein receptor (VLDLR), transcript variant 4, mRNA. (from RefSeq NM_001322226)
[MALRD1 \(ENST00000454679.7\) at chr10:19048801-19734478](#) - Homo sapiens MAM and LDL receptor class A domain containing 1 (MALRD1), mRNA. (from RefSeq NM_001142308)
[LRP8 \(ENST00000306052.12\) at chr1:53242364-53328070](#) - Homo sapiens LDL receptor related protein 8 (LRP8), transcript variant 1, mRNA. (from RefSeq NM_004631)
[LRP11 \(ENST00000239367.7\) at chr6:149818757-149864359](#) - Homo sapiens LDL receptor related protein 11 (LRP11), mRNA. (from RefSeq NM_032832)
[SORL1 \(ENST00000260197.12\) at chr11:121452314-121633763](#) - Homo sapiens sortilin related receptor 1 (SORL1), mRNA. (from RefSeq NM_003105)
[NR1H3 \(ENST00000407404.5\) at chr11:47248956-47268781](#) - Orphan receptor. Interaction with RXR shifts RXR from its role as a silent DNA-binding partner to an active ligand-
[LRP3 \(ENST00000253193.9\) at chr19:33194330-33208864](#) - Homo sapiens LDL receptor related protein 3 (LRP3), mRNA. (from RefSeq NM_002333)
[PCSK9 \(ENST00000302118.5\) at chr1:55039548-55064852](#) - Homo sapiens proprotein convertase subtilisin/kexin type 9 (PCSK9), transcript variant 1, mRNA. (from RefSeq NM_174936)
[MYLIP \(ENST00000356840.8\) at chr6:16129086-16148248](#) - Homo sapiens myosin regulatory light chain interacting protein (MYLIP), mRNA. (from RefSeq NM_013262)
[FCHO2 \(ENST00000430046.7\) at chr5:72956041-73090522](#) - Homo sapiens FCH and mu domain containing endocytic adaptor 2 (FCHO2), transcript variant 1, mRNA. (from RefSeq NM_138
[SNX17 \(ENST00000233575.7\) at chr2:27370616-27377535](#) - Homo sapiens sorting nexin 17 (SNX17), transcript variant 5, non-coding RNA. (from RefSeq NR_049782)
[LRP12 \(ENST00000276654.9\) at chr8:104489250-104589006](#) - Homo sapiens LDL receptor related protein 12 (LRP12), transcript variant 1, mRNA. (from RefSeq NM_013437)
[SCAP \(ENST00000265565.10\) at chr3:47413681-47475941](#) - Homo sapiens SREBF chaperone (SCAP), transcript variant 1, mRNA. (from RefSeq NM_012235)
[LRP10 \(ENST00000359591.9\) at chr14:22871740-22881713](#) - Homo sapiens LDL receptor related protein 10 (LRP10), transcript variant 1, mRNA. (from RefSeq NM_014045)
[LRP1B \(ENST00000389484.8\) at chr2:140231423-142131016](#) - Homo sapiens LDL receptor related protein 1B (LRP1B), mRNA. (from RefSeq NM_018557)
[PDLIM7 \(ENST00000355841.7\) at chr5:177483394-177497604](#) - Homo sapiens PDZ and LIM domain 7 (PDLIM7), transcript variant 1, mRNA. (from RefSeq NM_005451)
[DAB2 \(ENST00000320816.11\) at chr5:39371677-39424980](#) - Homo sapiens DAB adaptor protein 2 (DAB2), transcript variant 1, mRNA. (from RefSeq NM_001343)
[LRP4 \(ENST00000378623.6\) at chr11:46856717-46918550](#) - Homo sapiens LDL receptor related protein 4 (LRP4), mRNA. (from RefSeq NM_002334)
[LRP1 \(ENST00000243077.8\) at chr12:57128483-57213361](#) - Homo sapiens LDL receptor related protein 1 (LRP1), mRNA. (from RefSeq NM_002332)
[LRP2 \(ENST00000649046.1\) at chr2:169127109-169362534](#) - Homo sapiens LDL receptor related protein 2 (LRP2), mRNA. (from RefSeq NM_004525)
[GHR \(ENST00000230882.9\) at chr5:42423439-42721878](#) - Homo sapiens growth hormone receptor (GHR), transcript variant 1, mRNA. (from RefSeq NM_000163)
[ARRB2 \(ENST00000269260.7\) at chr17:4710632-4721497](#) - Homo sapiens arrestin beta 2 (ARRB2), transcript variant 1, mRNA. (from RefSeq NM_004313)
[LRP6 \(ENST00000261349.9\) at chr12:12116025-12267044](#) - Homo sapiens LDL receptor related protein 6 (LRP6), mRNA. (from RefSeq NM_002336)
[LRP5 \(ENST00000294304.12\) at chr11:68312591-68449275](#) - Homo sapiens LDL receptor related protein 5 (LRP5), transcript variant 1, mRNA. (from RefSeq NM_002335)
[MESD \(ENST00000561312.5\) at chr15:80947329-80989828](#) - Chaperone specifically assisting the folding of beta- propeller/EGF modules within the family of low-density lipoprot
[LDLRAP1 \(ENST00000374338.5\) at chr1:25543606-25568886](#) - Homo sapiens low density lipoprotein receptor adaptor protein 1 (LDLRAP1), mRNA. (from RefSeq NM_015627)
[LDLRAD1 \(ENST00000371360.2\) at chr1:54007298-54018186](#) - Homo sapiens low density lipoprotein receptor class A domain containing 1 (LDLRAD1), transcript variant 1, mRNA. (fr
[LDLRAD2 \(ENST00000344642.7\) at chr1:21812265-21825225](#) - Homo sapiens low density lipoprotein receptor class A domain containing 2 (LDLRAD2), mRNA. (from RefSeq NM_001013693
[LDLRAD3 \(ENST00000315571.6\) at chr11:35944062-36232136](#) - Homo sapiens low density lipoprotein receptor class A domain containing 3 (LDLRAD3), transcript variant 1, mRNA. (f
[LDLRAD4-AS1 \(ENST00000588672.1\) at chr18:13419421-13427480](#) - Homo sapiens LDLRAD4 antisense RNA 1 (LDLRAD4-AS1), long non-coding RNA. (from RefSeq NR_040031)
[LDLRAD4 \(ENST00000359446.10\) at chr18:13278105-13652754](#) - Homo sapiens low density lipoprotein receptor class A domain containing 4 (LDLRAD4), transcript variant 1, mRNA. (f
[AP2B1 \(ENST00000610402.5\) at chr17:35587322-35726413](#) - Homo sapiens adaptor related protein complex 2 subunit beta 1 (AP2B1), transcript variant 1, mRNA. (from RefSeq NM_00

[More results...](#)

Visual



Tracks

- Sets of data organized into categories
 - Categories: *Mapping and Sequencing, Genes and Gene Predictions, Phenotype and Literature, etc.*
- Each track is a display of a table of data
- The display of each track can be altered on a variety of parameters

Track Settings

GENCODE V36 Track Settings

genome.ucsc.edu/cgi-bin/hgTrackUi?hgsid=1064495105_sFNkAdpAAoXKGnMmJMO60eGRfE3J&c=chr19&g=knownGene

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GENCODE V36 Track Settings

GENCODE V36

([All Genes and Gene Predictions tracks](#))

Display mode: packSubmit

Label: ☒ Gene Symbol ☐ Ensembl ID ☐ UniProt display ID ☐ UCSC Genes ID

Show: ☒ non-coding genes ☒ splice variants ☐ pseudogenes

Tagged Sets: ☐ MANE only ☒ BASIC only ☐ All

Color track by codons: genomic codonsHelp on codon coloring

Show codon numbering: ☒

Display data as a density graph: ☐

[View table schema](#)

Data last updated at UCSC: 2021-01-14 15:32:12

Updated

Note: Uploaded by UCSC Mar. 17, 2020

Description

The GENCODE Genes track (version 36, Oct 2020) shows high-quality manual annotations merged with evidence-based automated annotations across the entire human genome generated by the [GENCODE project](#). By default, only the basic gene set is displayed, which is a subset of the comprehensive gene set. The basic set represents transcripts that GENCODE believes will be useful to the majority of users.

The track includes protein-coding genes, non-coding RNA genes, and pseudo-genes, though pseudo-genes are not displayed by default. It contains annotations on the reference chromosomes as well as assembly patches and alternative loci (haplotypes).

The following table provides statistics for the v36 release derived from the GTF file that contains annotations only on the main chromosomes. More information on how they were generated can be found in the [GENCODE site](#).

GENCODE v36 Release Stats			
Genes	Observed	Transcripts	Observed
Protein-coding genes	19,965	Protein-coding transcripts	83,986
Long non-coding RNA genes	17,910	- full length protein-coding	57,935
Small non-coding RNA genes	7,576	- partial length protein-coding	26,051
Pseudogenes	14,749	Nonsense mediated decay transcripts	15,811
Immunoglobulin/T-cell receptor gene segments	645	Long non-coding RNA loci transcripts	48,351

Underlying Table

GENCODE V36 Track Settings

genome.ucsc.edu/cgi-bin/hgTrackUi?hgsid=1064495105_sFNkAdpAAoXKGnMmJMO60eGRfE3J&c=chr19&g=knownGene

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GENCODE V36 Track Settings

GENCODE V36

[\(All Genes and Gene Predictions tracks\)](#)

Display mode: packSubmit

Label: ☒ Gene Symbol ☐ Ensembl ID ☐ UniProt display ID ☐ UCSC Genes ID

Show: ☒ non-coding genes ☒ splice variants ☐ pseudogenes

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Color track by codons: genomic codonsHelp on codon coloring

Show codon numbering: ☒

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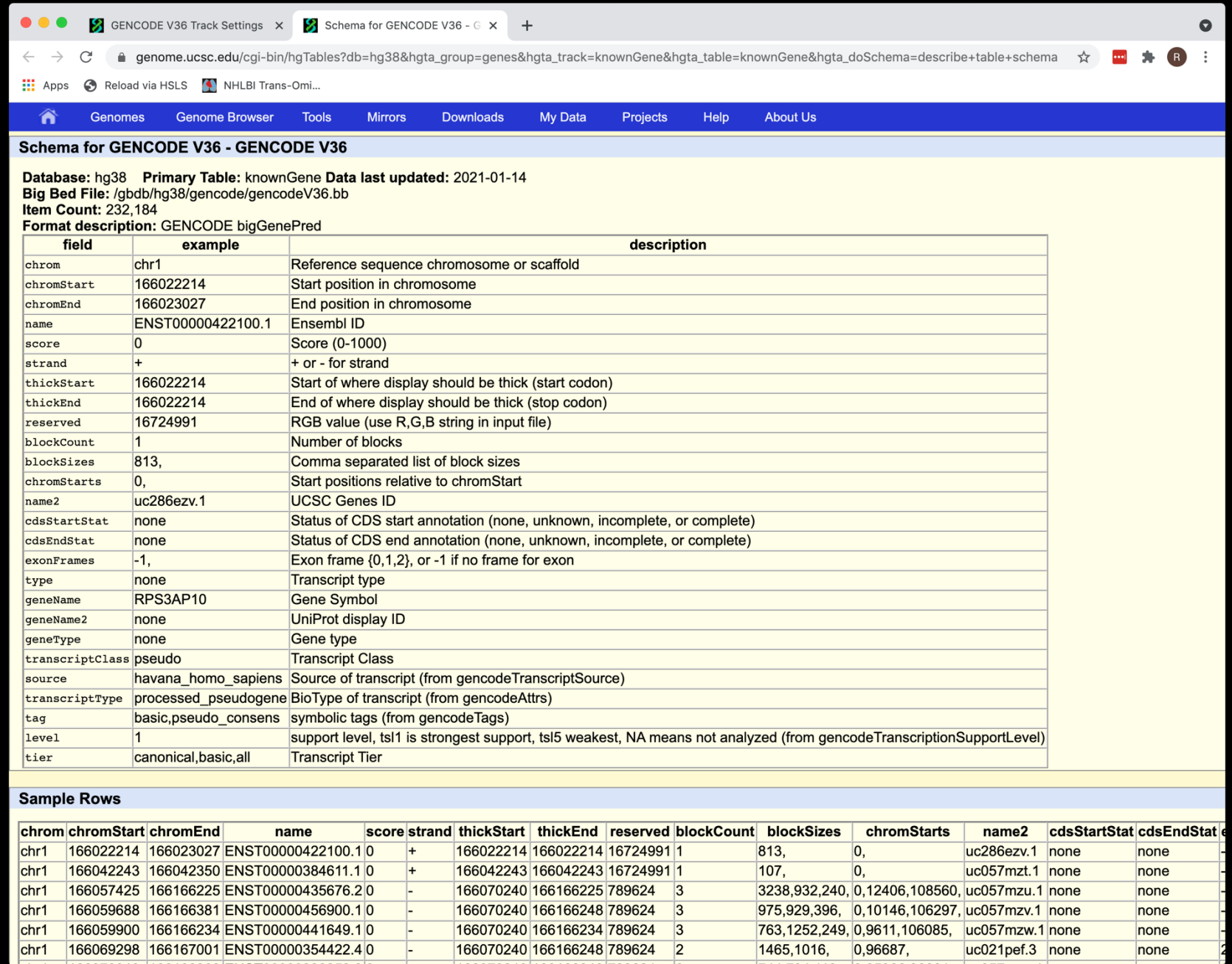
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Table Schema

- Schema = Design

- Note:
Database: hg38
Primary table:
knownGene
Fields: chrom,
chromStart, etc.
Example data



The screenshot shows a web browser window with the URL `genome.ucsc.edu/cgi-bin/hgTables?db=hg38&hgta_group=genes&hgta_track=knownGene&hgta_table=knownGene&hgta_doSchema=describe+table+schema`. The page title is "Schema for GENCODE V36 - GENCODE V36". The database is hg38, the primary table is knownGene, and the data was last updated on 2021-01-14. The Big Bed file is `/gdb/hg38/gencode/gencodeV36.bb` and the item count is 232,184. The format description is "GENCODE bigGenePred".

field	example	description
chrom	chr1	Reference sequence chromosome or scaffold
chromStart	166022214	Start position in chromosome
chromEnd	166023027	End position in chromosome
name	ENST00000422100.1	Ensembl ID
score	0	Score (0-1000)
strand	+	+ or - for strand
thickStart	166022214	Start of where display should be thick (start codon)
thickEnd	166022214	End of where display should be thick (stop codon)
reserved	16724991	RGB value (use R,G,B string in input file)
blockCount	1	Number of blocks
blockSizes	813,	Comma separated list of block sizes
chromStarts	0,	Start positions relative to chromStart
name2	uc286ezv.1	UCSC Genes ID
cdsStartStat	none	Status of CDS start annotation (none, unknown, incomplete, or complete)
cdsEndStat	none	Status of CDS end annotation (none, unknown, incomplete, or complete)
exonFrames	-1,	Exon frame {0,1,2}, or -1 if no frame for exon
type	none	Transcript type
geneName	RPS3AP10	Gene Symbol
geneName2	none	UniProt display ID
geneType	none	Gene type
transcriptClass	pseudo	Transcript Class
source	havana_homo_sapiens	Source of transcript (from gencodeTranscriptSource)
processed_pseudogene	none	BioType of transcript (from gencodeAttrs)
tag	basic,pseudo_consens	symbolic tags (from gencodeTags)
level	1	support level, tsl1 is strongest support, tsl5 weakest, NA means not analyzed (from gencodeTranscriptionSupportLevel)
tier	canonical,basic,all	Transcript Tier

Sample Rows

chrom	chromStart	chromEnd	name	score	strand	thickStart	thickEnd	reserved	blockCount	blockSizes	chromStarts	name2	cdsStartStat	cdsEndStat
chr1	166022214	166023027	ENST00000422100.1	0	+	166022214	166022214	16724991	1	813,	0,	uc286ezv.1	none	none
chr1	166042243	166042350	ENST00000384611.1	0	+	166042243	166042243	16724991	1	107,	0,	uc057mzt.1	none	none
chr1	166057425	166166225	ENST00000435676.2	0	-	166070240	166166225	789624	3	3238,932,240,	0,12406,108560,	uc057mzu.1	none	none
chr1	166059688	166166381	ENST00000456900.1	0	-	166070240	166166248	789624	3	975,929,396,	0,10146,106297,	uc057mzv.1	none	none
chr1	166059900	166166234	ENST00000441649.1	0	-	166070240	166166234	789624	3	763,1252,249,	0,9611,106085,	uc057mzw.1	none	none
chr1	166069298	166167001	ENST00000354422.4	0	-	166070240	166166248	789624	2	1465,1016,	0,96687,	uc021pef.3	none	none

A Different Table Schema

- Go back to Genome View and click on dbSNP 153
- Schema links are at the far right of the subtracks

dbSnp153Common

[illegible]

Features

- Visual Interface ("**Genome Browser**" on the landing page)
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Table Browser

dbSNP 153 Track Settings

Table Browser

genome.ucsc.edu/cgi-bin/hgTables?hgsid=1064509817_bNg5WVEhT4U3SDf7psSjadBBzg8n

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Table Browser

Use this program to retrieve the data associated with a track in text format, to calculate intersections between tracks, and to retrieve DNA sequence covered by a track. For help in using this application see [Using the Table Browser](#) for a description of the controls in this form, and the [User's Guide](#) for general information and sample queries. For more complex queries, you may want to use [Galaxy](#) or our [public MySQL server](#). To examine the biological function of your set through annotation enrichments, send the data to [GREAT](#). Refer to the [Credits](#) page for the list of contributors and usage restrictions associated with these data. All tables can be downloaded in their entirety from the [Sequence and Annotation Downloads](#) page.

clade: **genome:** **assembly:**

group: **track:**

table:

Note: Most dbSNP tables are huge. Trying to download them through the Table Browser usually leads to a timeout. Please see our [Data Access FAQ](#) on how to download dbSNP data.

region: ☐ genome ☒ position

identifiers (names/accessions):

filter:

subtrack merge:

intersection:

output format: Send output to ☐ [Galaxy](#) ☐ [GREAT](#)

output file: (leave blank to keep output in browser)

file type returned: ☒ plain text ☐ gzip compressed

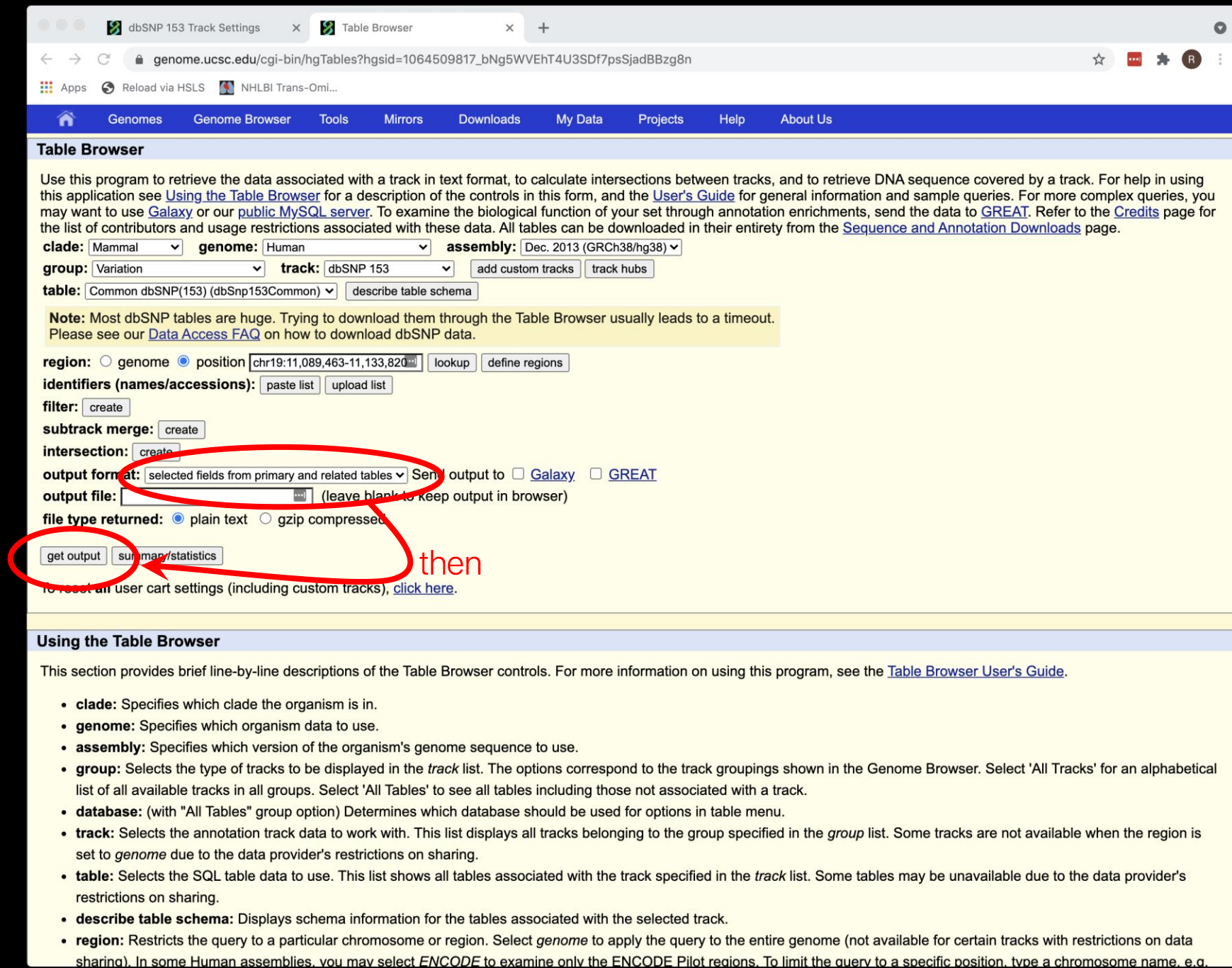
To reset **all** user cart settings (including custom tracks), [click here](#).

Using the Table Browser

This section provides brief line-by-line descriptions of the Table Browser controls. For more information on using this program, see the [Table Browser User's Guide](#).

- **clade:** Specifies which clade the organism is in.
- **genome:** Specifies which organism data to use.
- **assembly:** Specifies which version of the organism's genome sequence to use.
- **group:** Selects the type of tracks to be displayed in the *track* list. The options correspond to the track groupings shown in the Genome Browser. Select 'All Tracks' for an alphabetical list of all available tracks in all groups. Select 'All Tables' to see all tables including those not associated with a track.
- **database:** (with "All Tables" group option) Determines which database should be used for options in table menu.
- **track:** Selects the annotation track data to work with. This list displays all tracks belonging to the group specified in the *group* list. Some tracks are not available when the region is set to *genome* due to the data provider's restrictions on sharing.
- **table:** Selects the SQL table data to use. This list shows all tables associated with the track specified in the *track* list. Some tables may be unavailable due to the data provider's restrictions on sharing.
- **describe table schema:** Displays schema information for the tables associated with the selected track.
- **region:** Restricts the query to a particular chromosome or region. Select *genome* to apply the query to the entire genome (not available for certain tracks with restrictions on data sharing). In some Human assemblies, you may select *ENCODE* to examine only the ENCODE Pilot regions. To limit the query to a specific position, type a chromosome name, e.g.,

Select fields



dbSNP 153 Track Settings | Table Browser

genome.ucsc.edu/cgi-bin/hgTables?hgsid=1064509817_bNg5WVEhT4U3SDf7psSjadBBzg8n

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clade: genome: assembly:

group: track:

table:

Note: Most dbSNP tables are huge. Trying to download them through the Table Browser usually leads to a timeout. Please see our [Data Access FAQ](#) on how to download dbSNP data.

region: ☐ genome ☒ position

identifiers (names/accessions):

filter:

subtrack merge:

intersection:

output format: Send output to ☐ [Galaxy](#) ☐ [GREAT](#)

output file: (leave blank to keep output in browser)

file type returned: ☒ plain text ☐ gzip compressed

then

to reset all user cart settings (including custom tracks), [click here](#).

Using the Table Browser

This section provides brief line-by-line descriptions of the Table Browser controls. For more information on using this program, see the [Table Browser User's Guide](#).

- **clade:** Specifies which clade the organism is in.
- **genome:** Specifies which organism data to use.
- **assembly:** Specifies which version of the organism's genome sequence to use.
- **group:** Selects the type of tracks to be displayed in the *track* list. The options correspond to the track groupings shown in the Genome Browser. Select 'All Tracks' for an alphabetical list of all available tracks in all groups. Select 'All Tables' to see all tables including those not associated with a track.
- **database:** (with "All Tables" group option) Determines which database should be used for options in table menu.
- **track:** Selects the annotation track data to work with. This list displays all tracks belonging to the group specified in the *group* list. Some tracks are not available when the region is set to *genome* due to the data provider's restrictions on sharing.
- **table:** Selects the SQL table data to use. This list shows all tables associated with the track specified in the *track* list. Some tables may be unavailable due to the data provider's restrictions on sharing.
- **describe table schema:** Displays schema information for the tables associated with the selected track.
- **region:** Restricts the query to a particular chromosome or region. Select *genome* to apply the query to the entire genome (not available for certain tracks with restrictions on data sharing). In some Human assemblies, you may select *ENCODE* to examine only the ENCODE Pilot regions. To limit the query to a specific position, type a chromosome name, e.g.

Select fields

dbSNP 153 Track Settings

Select Fields from hg38.dbSnp

genome.ucsc.edu/cgi-bin/hgTables

AppsReload via HSLSNHLBI Trans-Omi...

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Select Fields from hg38.dbSnp153Common

☐	chrom	Reference sequence chromosome or scaffold
☐	chromStart	Start position in chrom
☐	chromEnd	End position in chrom
☐	name	dbSNP Reference SNP (rs) identifier
☐	ref	Reference allele; usually major allele, but may be minor allele
☐	altCount	Number of alternate alleles (usually 1)
☐	alts	Alternate (non-reference) alleles; may include major allele
☐	shiftBases	Bases by which an indel placement could be shifted left or right (display shows thin line over uncertain region, shifts minimal representation right)
☐	freqSourceCount	Number of projects reporting frequencies in current dbSNP build
☐	minorAlleleFreq	Minor allele frequency, i.e. second highest allele frequency, from each frequency source; NaN if no data from project
☐	majorAllele	Allele most frequently observed by each source
☐	minorAllele	Allele second most frequently observed by each source
☐	maxFuncImpact	Sequence Ontology (SO) ID number for greatest functional impact on gene; 0 if no SO terms are annotated
☐	class	Variation class/type
☐	ucscNotes	Interesting or anomalous properties noted by UCSC
☐	_dataOffset	Offset into bigDbSnpDetails file for line with more info
☐	_dataLen	Length of line in bigDbSnpDetails

get output

cancel

check all

clear all

Filter

dbSNP 153 Track Settings

Table Browser

genome.ucsc.edu/cgi-bin/hgTables?hgsid=1064509817_bNg5WVEhT4U3SDf7psSjadBBzg8n

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Table Browser

Use this program to retrieve the data associated with a track in text format, to calculate intersections between tracks, and to retrieve DNA sequence covered by a track. For help in using this application see [Using the Table Browser](#) for a description of the controls in this form, and the [User's Guide](#) for general information and sample queries. For more complex queries, you may want to use [Galaxy](#) or our [public MySQL server](#). To examine the biological function of your set through annotation enrichments, send the data to [GREAT](#). Refer to the [Credits](#) page for the list of contributors and usage restrictions associated with these data. All tables can be downloaded in their entirety from the [Sequence and Annotation Downloads](#) page.

clade: genome: assembly:

group: track:

table:

Note: Most dbSNP tables are huge. Trying to download them through the Table Browser usually leads to a timeout. Please see our [Data Access FAQ](#) on how to download dbSNP data.

region: ☐ genome ☒ position

identifiers (names/accessions):

filter:

subtrack merge:

intersection:

output format: Send output to ☐ [Galaxy](#) ☐ [GREAT](#)

output file: (leave blank to keep output in browser)

file type returned: ☒ plain text ☐ gzip compressed

To reset all user cart settings (including custom tracks), [click here](#).

Using the Table Browser

This section provides brief line-by-line descriptions of the Table Browser controls. For more information on using this program, see the [Table Browser User's Guide](#).

- clade:** Specifies which clade the organism is in.
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- group:** Selects the type of tracks to be displayed in the *track* list. The options correspond to the track groupings shown in the Genome Browser. Select 'All Tracks' for an alphabetical list of all available tracks in all groups. Select 'All Tables' to see all tables including those not associated with a track.
- database:** (with "All Tables" group option) Determines which database should be used for options in table menu.
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- table:** Selects the SQL table data to use. This list shows all tables associated with the track specified in the *track* list. Some tables may be unavailable due to the data provider's restrictions on sharing.
- describe table schema:** Displays schema information for the tables associated with the selected track.
- region:** Restricts the query to a particular chromosome or region. Select *genome* to apply the query to the entire genome (not available for certain tracks with restrictions on data sharing). In some Human assemblies, you may select *ENCODE* to examine only the ENCODE Pilot regions. To limit the query to a specific position, type a chromosome name, e.g.

Filter

dbSNP 153 Track Settings

Filter on Fields from hg38.dbS

genome.ucsc.edu/cgi-bin/hgTables

AppsReload via HSLSNHLBI Trans-Omi...

GenomesGenome BrowserToolsMirrorsDownloadsMy DataProjectsHelpAbout Us

Filter on Fields from hg38.dbSnp153Common

chrom

does

match

*

chromStart

is

ignored

0

AND

chromEnd

is

ignored

0

AND

name

does

match

*

AND

ref

does

match

*

altCount

is

ignored

0

AND

alts

does

match

*

shiftBases

is

ignored

0

AND

freqSourceCount

is

ignored

0

AND

minorAlleleFreq

does

match

*

majorAllele

does

match

*

minorAllele

does

match

*

maxFuncImpact

is

ignored

0

AND

class

does

match

☒ *

☐ snv

☐ mnv

☐ ins

☐ del

☐ delins

☐ identity

AND

ucscNotes

does

match

*

_dataOffset

is

ignored

0

AND

_dataLen

is

ignored

0

AND

AND

Free-form query:

submit

cancel

Output

https://genome.ucsc.edu/cgi-bin/hgTables				
genome.ucsc.edu/cgi-bin/hgTables				
Apps Reload via HSLs NHLBI Trans-Omi...				
#chrom	chromStart	chromEnd	name	
chr19	11089632	11089633	rs17242759	
chr19	11089735	11089736	rs12981050	
chr19	11089754	11089755	rs143643606	
chr19	11089798	11089799	rs17242766	
chr19	11090129	11090130	rs17242353	
chr19	11090447	11090448	rs57217136	
chr19	11091312	11091315	rs60173709	
chr19	11091518	11091522	rs141787760	
chr19	11091629	11091630	rs6511720	
chr19	11091839	11091840	rs8104549	
chr19	11091892	11091893	rs8104576	
chr19	11092737	11092738	rs17249169	
chr19	11093673	11093674	rs13345161	
chr19	11093771	11093772	rs77238163	
chr19	11093885	11093886	rs76058149	
chr19	11093950	11093951	rs17242367	
chr19	11094139	11094140	rs17249176	
chr19	11094206	11094207	rs143437024	
chr19	11094305	11094306	rs10425654	
chr19	11094323	11094324	rs17249190	
chr19	11094460	11094461	rs118183202	
chr19	11094680	11094681	rs17242822	
chr19	11095298	11095299	rs8102912	
chr19	11095853	11095854	rs8110695	
chr19	11095898	11095899	rs6511721	
chr19	11095908	11095909	rs74516607	
chr19	11096009	11096010	rs74489414	
chr19	11096052	11096053	rs17242381	
chr19	11096292	11096293	rs17242395	
chr19	11096425	11096426	rs1010679	
chr19	11096780	11096781	rs76987683	
chr19	11096839	11096840	rs73015030	
chr19	11096994	11096995	rs117972228	
chr19	11097100	11097101	rs117423069	
chr19	11097305	11097306	rs67337506	
chr19	11097403	11097404	rs116436750	
chr19	11098795	11098796	rs149564349	
chr19	11098858	11098859	rs74938451	
chr19	11098899	11098900	rs17248769	
chr19	11098952	11098953	rs73015033	
chr19	11098993	11099002	rs142444132	
chr19	11099045	11099046	rs73015034	
chr19	11099081	11099083	rs141820146	
chr19	11099087	11099088	rs74857287	
chr19	11099312	11099333	rs59029506	
chr19	11099313	11099314	rs560115066	
chr19	11099327	11099328	rs8111982	
chr19	11099577	11099578	rs17248776	
chr19	11099637	11099638	rs17248783	
chr19	11100235	11100236	rs2228671	
chr19	11100244	11100245	rs72658855	
chr19	11100400	11100401	rs3745677	
chr19	11100402	11100403	rs3745678	
chr19	11100531	11100532	rs72658856	
chr19	11100793	11100794	rs371464789	
chr19	11101502	11101503	rs28493980	
chr19	11101793	11101794	rs12985460	
chr19	11102035	11102036	rs59078413	
chr19	11102086	11102087	rs36096887	
chr19	11102264	11102265	rs116530815	
chr19	11103012	11103013	rs2304183	
chr19	11103798	11103799	rs74257940	
chr19	11103856	11103857	rs560550	

Features

- Visual Interface ("**Genome Browser**" on the landing page)
- Underlying Data Tables ("**Table Browser**")
- Two types of data integrators
 - **Site Based** ("**Variant Annotation Integrator**")
 - **Range Based** ("**Data Integrator**")
- Desktop version [**"Genome Browser in a Box (GBiB)"**]
- Genome coordinate conversion ("**LiftOver**")
- Ability to add custom tracks ("**Track Hubs**")

- Site based

Variant Annotation Integrator

genome.ucsc.edu/cgi-bin/hgVai?hgsid=1064525655_FsOGPI0csyxuLmSVipmPYmKxx4mQ

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Variant Annotation Integrator

Select Genome Assembly and Region

cladegenomeassembly
MammalHumanDec. 2013 (GRCh38/hg38)
region to annotate
genome

Select Variants

Your session doesn't have any custom tracks or hub tracks in [pgSnp](#) or [VCF](#) format. [add pgSnp or VCF custom track](#) [add track hub](#)

variants: Artificial Example Variants
maximum number of variants to be processed: 10,000
[add custom tracks](#) [track hubs](#) To reset all user cart settings (including custom tracks), [click here](#).

Select Genes

The gene predictions selected here will be used to determine the effect of each variant on genes, for example intronic, missense, splice site, intergenic etc.
GENCODE V36

Select Regulatory Annotations

The annotations in this section provide predicted regulatory regions based on various experimental data. When a variant overlaps an annotation selected here, the consequence term [regulatory_region_variant](#) will be assigned. Follow the links to description pages that explain how each dataset was constructed. Some datasets cover a significant portion of the genome and it may be desirable to filter these annotations by cell type and/or score in order to avoid an overabundance of hits.
☐ [DNase I Hypersensitivity Peak Clusters from ENCODE \(95 cell types\)](#)
[+ filter items](#)

Select More Annotations (optional)

☐ **Database of Non-synonymous Functional Predictions (dbNSFP)**
[dbNSFP \(Liu et al. 2015\)](#) release 3.1a provides pre-computed scores and predictions of functional significance from a variety of tools. Every possible coding change to transcripts in Gencode release 22 (Ensembl 79, Mar. 2015) gene predictions has been evaluated. *Note: This may not encompass all transcripts in your selected gene set.*
[Set all](#) [Clear all](#)
☐ [Variant Effect Scoring Tool \(VEST\)](#) (scores [0-1] predict confidence that a change is deleterious)
☒ [SIFT](#) (D = damaging, T = tolerated)
☒ [PolyPhen-2](#) with HumDiv training set (D = probably damaging, P = possibly damaging, B = benign)
☒ [PolyPhen-2](#) with HumVar training set (D = probably damaging, P = possibly damaging, B = benign)
☐ [MutationTaster](#) (A = disease causing automatic, D = disease causing, N = polymorphism, P = polymorphism automatic)
☐ [MutationAssessor](#) (high or medium: predicted functional; low or neutral: predicted non-functional)
☐ [Likelihood ratio test \(LRT\)](#) (D = deleterious, N = Neutral, U = unknown)
☐ [InterPro](#) protein domains
☐ [GERP++](#) Rejected Substitutions (RS)
☐ [GERP++](#) Neutral Rate (NR)

[+ Transcript status](#)
☐ **HGVS variant nomenclature**
The [Human Genome Variation Society \(HGVS\)](#) has established a [sequence variant nomenclature](#), an international standard used to report variation in genomic, transcript and protein

Output

https://genome.ucsc.edu/cgi-bin/hgVai

genome.ucsc.edu/cgi-bin/hgVai

Apps Reload via HSLS NHLBI Trans-Omi...

```

## ENSEMBL VARIANT EFFECT PREDICTOR format (UCSC Variant Annotation Integrator)
## Output produced at 2021-03-23 07:15:55
## Connected to UCSC database hg38
## Variants: Artificial Example Variants
## Transcripts: GENCODE V36 (/gdb/hg38/gencode/gencodeV36.bb)
## dbSNP: Simple Nucleotide Polymorphisms (dbSNP 151) (/gdb/hg38/vai/snp151.bed4.bb)
## Keys for Extra column items:
## SIFT: (http://sift.bii.a-star.edu.sg/) SIFT (D = damaging, T = tolerated)
## PP2HVAR: (http://genetics.bwh.harvard.edu/pph2/) PolyPhen-2 with HumVar training set (D = probably damaging, P = possibly damaging, B = benign)
## PP2HDIV: (http://genetics.bwh.harvard.edu/pph2/) PolyPhen-2 with HumDiv training set (D = probably damaging, P = possibly damaging, B = benign)
Uploaded Variation Location Allele Gene Feature Feature type Consequence Position in cDNA Position in CDS Position in protein Amino acid
change Codon change Co-located Variation Extra
ex_igLeft chr19:11084457 C - - intergenic_variant - - - - - -
ex_updownLeft chr19:11089461 A uc060toi.1 ENST00000557933.5 Transcript upstream_gene_variant - - - - - rs879254376
DISTANCE=1
ex_updownLeft chr19:11089461 A uc060toj.1 ENST00000560502.5 Transcript upstream_gene_variant - - - - - rs879254376
DISTANCE=2
ex_updownLeft chr19:11089461 A uc060tok.1 ENST00000557958.1 Transcript upstream_gene_variant - - - - - rs879254376
DISTANCE=2
ex_updownLeft chr19:11089461 A uc010xlo.3 ENST00000455727.6 Transcript upstream_gene_variant - - - - - rs879254376
DISTANCE=2
ex_updownLeft chr19:11089461 A uc010xll.3 ENST00000535915.5 Transcript upstream_gene_variant - - - - - rs879254376
DISTANCE=2
ex_updownLeft chr19:11089461 A uc010xln.3 ENST00000545707.5 Transcript upstream_gene_variant - - - - - rs879254376
DISTANCE=2
ex_updownLeft chr19:11089461 A uc002mqk.6 ENST00000558518.6 Transcript upstream_gene_variant - - - - - rs879254376
DISTANCE=2
ex_updownLeft chr19:11089461 A uc010xlk.3 ENST00000558013.5 Transcript upstream_gene_variant - - - - - rs879254376
DISTANCE=17
ex_updownLeft chr19:11089461 A uc060tol.1 ENST00000252444.9 Transcript upstream_gene_variant - - - - - rs879254376
DISTANCE=1138
ex_utrLeft chr19:11089463 A uc060toi.1 ENST00000557933.5 Transcript 5_prime_UTR_variant 2 - - - -
EXON=1/18
ex_utrLeft chr19:11089463 A uc060toj.1 ENST00000560502.5 Transcript non_coding_transcript_exon_variant 1 - - - -
- EXON=1/2
ex_utrLeft chr19:11089463 A uc060tok.1 ENST00000557958.1 Transcript non_coding_transcript_exon_variant 1 - - - -
- EXON=1/3
ex_utrLeft chr19:11089463 A uc010xlo.3 ENST00000455727.6 Transcript 5_prime_UTR_variant 1 - - - -
EXON=1/16
ex_utrLeft chr19:11089463 A uc010xll.3 ENST00000535915.5 Transcript 5_prime_UTR_variant 1 - - - -
EXON=1/17
ex_utrLeft chr19:11089463 A uc010xln.3 ENST00000545707.5 Transcript 5_prime_UTR_variant 1 - - - -
EXON=1/16
ex_utrLeft chr19:11089463 A uc002mqk.6 ENST00000558518.6 Transcript 5_prime_UTR_variant 1 - - - -
EXON=1/18
ex_utrLeft chr19:11089463 A uc010xlk.3 ENST00000558013.5 Transcript upstream_gene_variant - - - - -
DISTANCE=15
ex_utrLeft chr19:11089463 A uc060tol.1 ENST00000252444.9 Transcript upstream_gene_variant - - - - -
DISTANCE=1136
ex_nce chr19:11089539 C uc060toi.1 ENST00000557933.5 Transcript 5_prime_UTR_variant 78 - - - - EXON=1/18
ex_nce chr19:11089539 C uc060toj.1 ENST00000560502.5 Transcript non_coding_transcript_exon_variant 77 - - - -
EXON=1/2
ex_nce chr19:11089539 C uc060tok.1 ENST00000557958.1 Transcript non_coding_transcript_exon_variant 77 - - - -
EXON=1/3
ex_nce chr19:11089539 C uc010xlo.3 ENST00000455727.6 Transcript 5_prime_UTR_variant 77 - - - - EXON=1/16
ex_nce chr19:11089539 C uc010xll.3 ENST00000535915.5 Transcript 5_prime_UTR_variant 77 - - - - EXON=1/17
ex_nce chr19:11089539 C uc010xln.3 ENST00000545707.5 Transcript 5_prime_UTR_variant 77 - - - - EXON=1/16
ex_nce chr19:11089539 C uc002mqk.6 ENST00000558518.6 Transcript 5_prime_UTR_variant 77 - - - - EXON=1/18
ex_nce chr19:11089539 C uc010xlk.3 ENST00000558013.5 Transcript 5_prime_UTR_variant 62 - - - - EXON=1/18
ex_nce chr19:11089539 C uc060tol.1 ENST00000252444.9 Transcript upstream_gene_variant - - - - -
DISTANCE=1060
ex_codonLeftDel chr19:11089551-11089553 - uc060toi.1 ENST00000557933.5 Transcript inframe_deletion 91-93 4-6 2 G/- GGG/- -
EXON=1/18

```

Features

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Data Integrator

- Range based

Data Integrator

genome.ucsc.edu/cgi-bin/hgIntegrator

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Data Integrator

UndoRedo

Select Genome Assembly and Region

group

genome

assembly

Mammal

Human

Dec. 2013 (GRCh38/hg38)

region to annotate

position or search term

chr19:11,089,463-11,133,820

Configure Data Sources

please add at least one data source

Add Data Source

track group

track

Mapping and Sequencing

Fix Patches (fixSeqLiftOverPsl)

View table schema

Add

get more data:

track hubs

custom tracks

Output Options

At least one data source must be added.

Using the Data Integrator

The Data Integrator finds items in different tracks that overlap by position, and unlike the Table Browser's intersection function, the Data Integrator can output all fields from all selected tracks. Up to 5 different tracks may be queried at a time.

This section contains a brief overview of Data Integrator controls. For more information on using the tools, see the [Data Integrator User's Guide](#).

Select Genome Assembly and Region

The controls in this section are for selecting a genome assembly and region to search.

- **group**: A species group: Mammal, Vertebrate, Insect etc.
- **genome**: A single species such as Human or Mouse (not available for certain tracks with restrictions on data sharing).
- **assembly**: A version of the reference genome assembly such as GRCh37/hg19.

Configure Data Sources

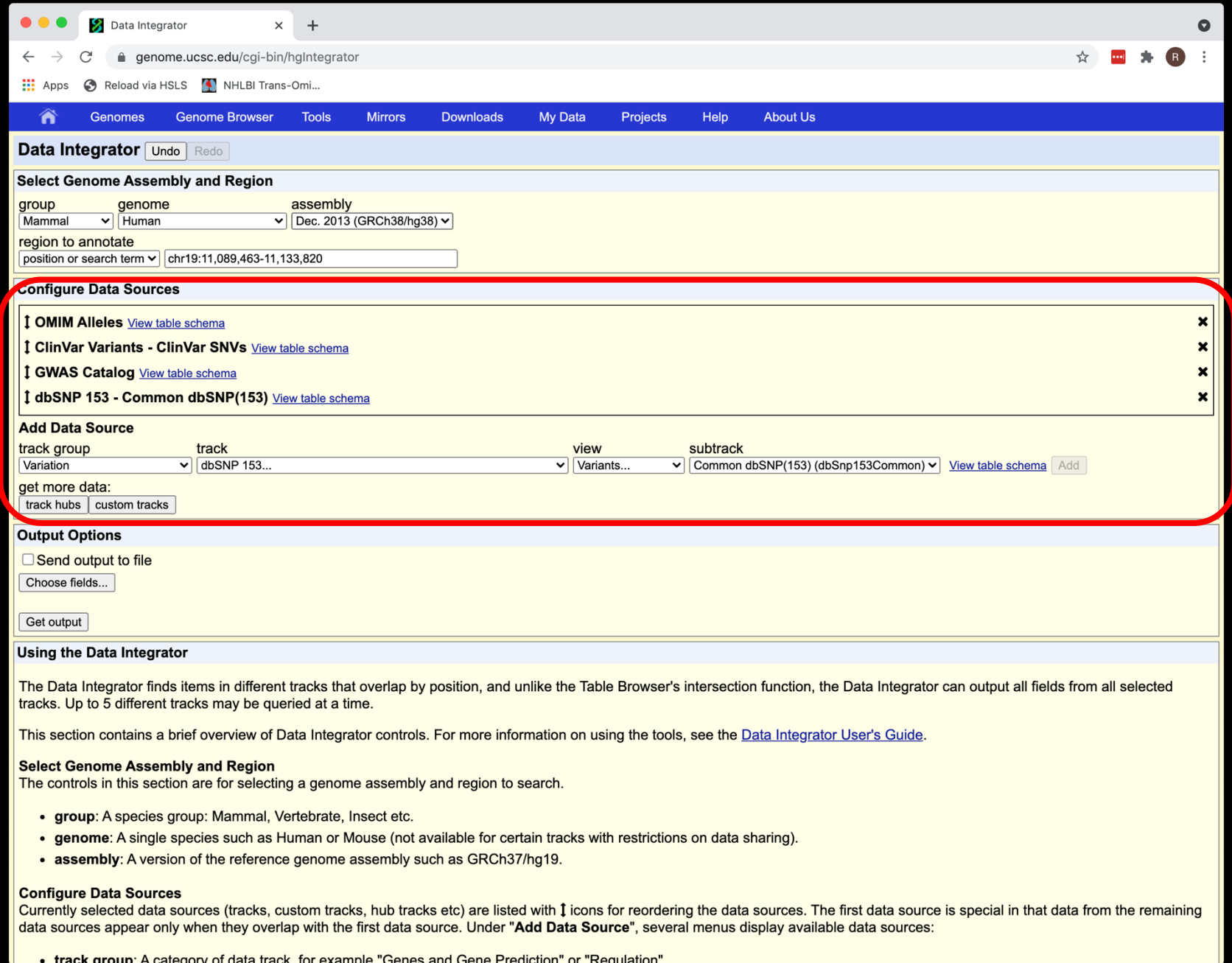
Currently selected data sources (tracks, custom tracks, hub tracks etc) are listed with  icons for reordering the data sources. The first data source is special in that data from the remaining data sources appear only when they overlap with the first data source. Under "**Add Data Source**", several menus display available data sources:

- **track group**: A category of data track, for example "Genes and Gene Prediction" or "Regulation".
- **track**: One or more data tables containing results of an experiment or a group of closely related experiments. Some tracks are not available when the region is set to **genome** due to the data provider's restrictions on sharing.
- **table**: This appears only when the selected track has more than one data table.

These sections can be reordered by dragging on the section title or arrow icon on the left. To remove a section, click the  icon to the right of the title. Click on the Add button to add a new data source.

Output Options

Add tracks



Data Integrator Undo Redo

Select Genome Assembly and Region

group genome assembly
Mammal Human Dec. 2013 (GRCh38/hg38)
region to annotate
position or search term chr19:11,089,463-11,133,820

Configure Data Sources

- ↓ OMIM Alleles [View table schema](#)
- ↓ ClinVar Variants - ClinVar SNVs [View table schema](#)
- ↓ GWAS Catalog [View table schema](#)
- ↓ dbSNP 153 - Common dbSNP(153) [View table schema](#)

Add Data Source

track group track view subtrack
Variation dbSNP 153... Variants... Common dbSNP(153) (dbSnp153Common) [View table schema](#) Add

get more data:
track hubs custom tracks

Output Options

☐ Send output to file
[Choose fields...](#)
[Get output](#)

Using the Data Integrator

The Data Integrator finds items in different tracks that overlap by position, and unlike the Table Browser's intersection function, the Data Integrator can output all fields from all selected tracks. Up to 5 different tracks may be queried at a time.

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The controls in this section are for selecting a genome assembly and region to search.

- **group**: A species group: Mammal, Vertebrate, Insect etc.
- **genome**: A single species such as Human or Mouse (not available for certain tracks with restrictions on data sharing).
- **assembly**: A version of the reference genome assembly such as GRCh37/hg19.

Configure Data Sources
Currently selected data sources (tracks, custom tracks, hub tracks etc) are listed with ↓ icons for reordering the data sources. The first data source is special in that data from the remaining data sources appear only when they overlap with the first data source. Under "Add Data Source", several menus display available data sources:

- **track group**: A category of data track, for example "Genes and Gene Prediction" or "Regulation".

Output

[illegible]

Genome Browser Creating Custom Tracks

HUGEN 2073

Genomic Data Visualization & Integration

Slides adapted from Ryan L. Minster, PhD, MSIS

Learning Objectives

- By the end of the session, students will be able to:
 - Describe the various coordinate systems used in genomic data and will be able to convert between them
 - Describe and use the RGB color system to assign colors to data
 - Create a track for the UCSC Genome Browser in BED format
 - Create a track for the UCSC Genome Browser in bedGraph format
 - Create a track for the UCSC Genome Browser in Personal Genome SNP format

Chromosome Names

- b37 Convention
(1000 Genomes Project)

1

2

⋮

22

X

Y

MT

- hg19 Convention
(UCSC Genome Browser)

chr1

chr2

⋮

chr22

chrX

chrY

chrM

Also, the current convention of the
Genome Reference Consortium

Range Conventions

- 1-start, fully-closed

A	C	G	T	A
1	2	3	4	5

chr1:1-5

.gtf

.gff

.sam

.vcf

.wig

HGVS Nomenclature

- 0-start, half-open

A	C	G	T	A	
0	1	2	3	4	5

chr1 0 5

*Helps to think of it as
counting between the bases.*

.bed (not PLINK .bed)

.bam

.bcf

Range Conventions

- 1-start, fully-closed

A	C	G	T	A
1	2	3	4	5

g.2del

chr1:2-2

g.1_2insA

chr1:1-2

g.3G>T

chr1:3-3

- 0-start, half-open

A	C	G	T	A	
0	1	2	3	4	5

g.2del

chr1 1 2

g.1_2insA

chr1 1 1

g.3G>T

chr1 2 3

Range Conventions

- Why 0-start? Because the math is easier.

seq1		A	C	G	T	A					
seq2					T	A	G	C	T	T	
1-based		23	24	25	26	27	28	29	30	31	
0-based	22	23	24	25	26	27	28	29	30	31	

- What is the length of the segment from A to A of seq1?

$$\text{1-based } 27 - 23 + 1 = 5$$

$$\text{0-based } 27 - 22 = 5$$

- What is the overlap length of seq1 and seq2?

$$\text{1-based } 27 - 26 + 1 = 2$$

$$\text{0-based } 27 - 25 = 2$$

Converting from 1-start to 0-start

- If the feature has a length > 0, subtract 1 from the start coordinate and leave the end coordinate the same

An exon starting with 10 and ending with 200:

chr1:10-200 -> chr1 9 200

A SNP at nucleotide 100 (i.e., starting with 100 and ending with 100):

chr1:100-100 -> chr1 99 100

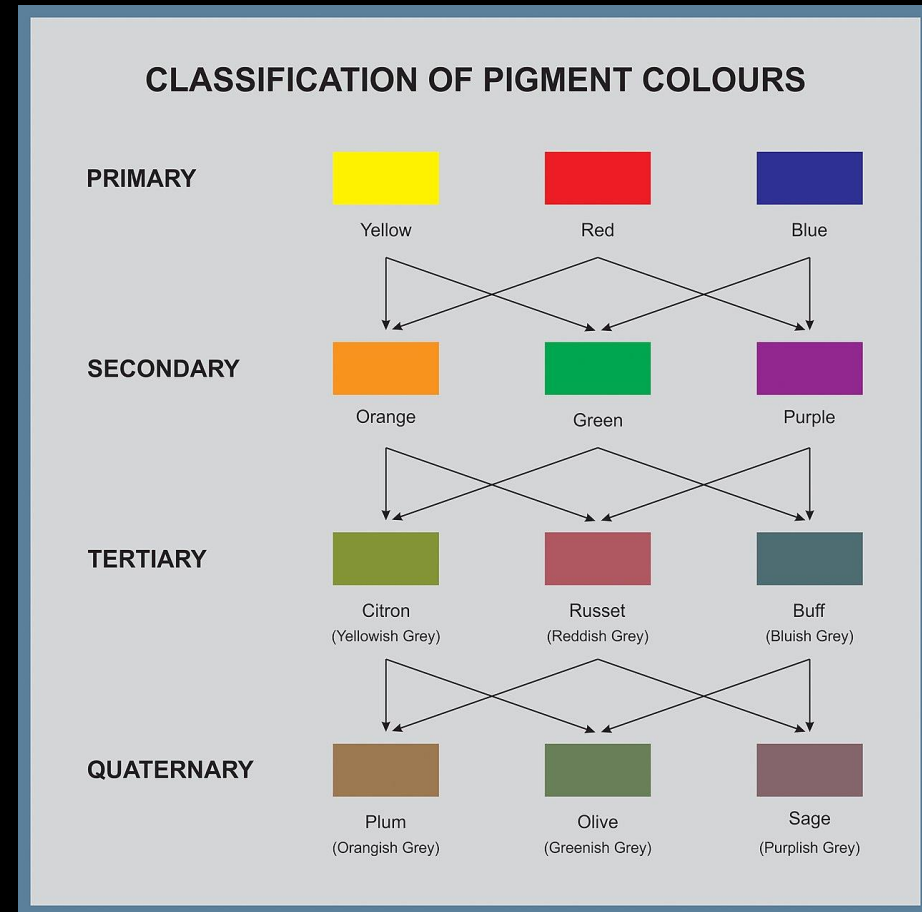
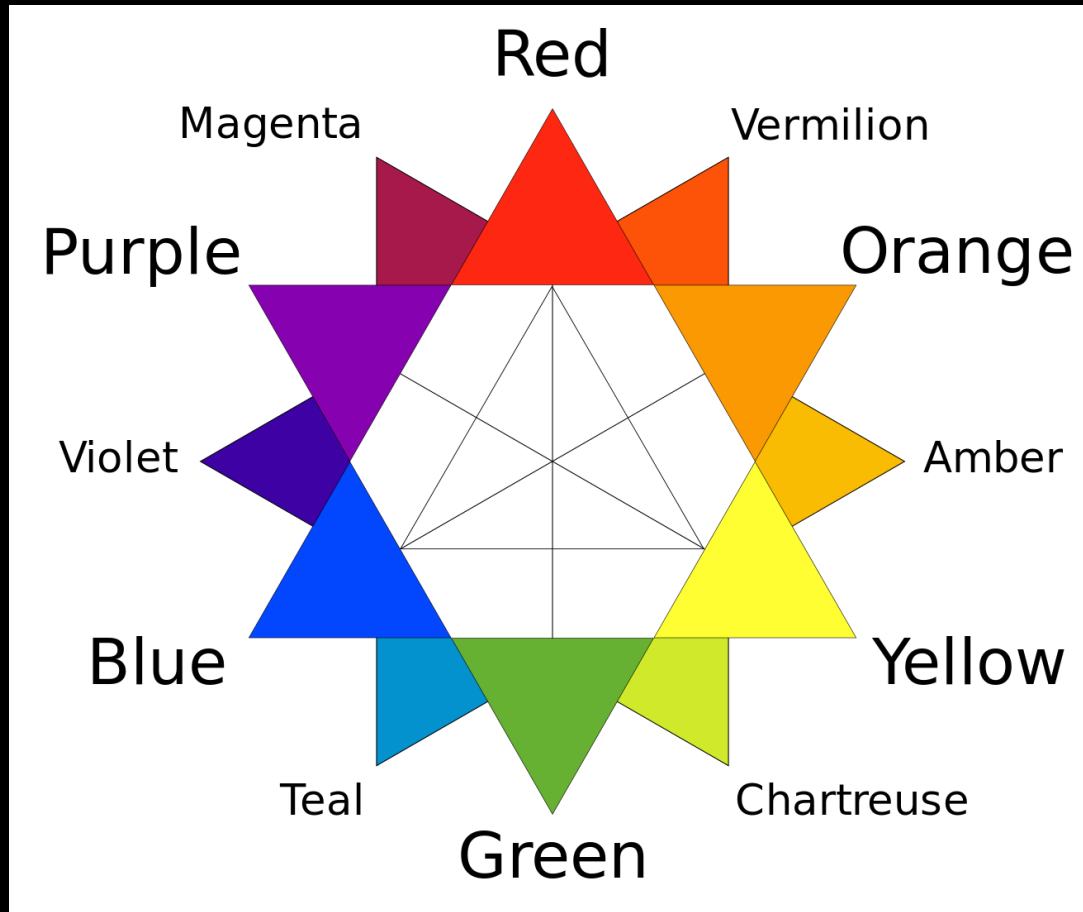
Converting from 1-start to 0-start

- If the feature has a length = 0 (an insertion), leave the start coordinate the same and subtract 1 from the end coordinate

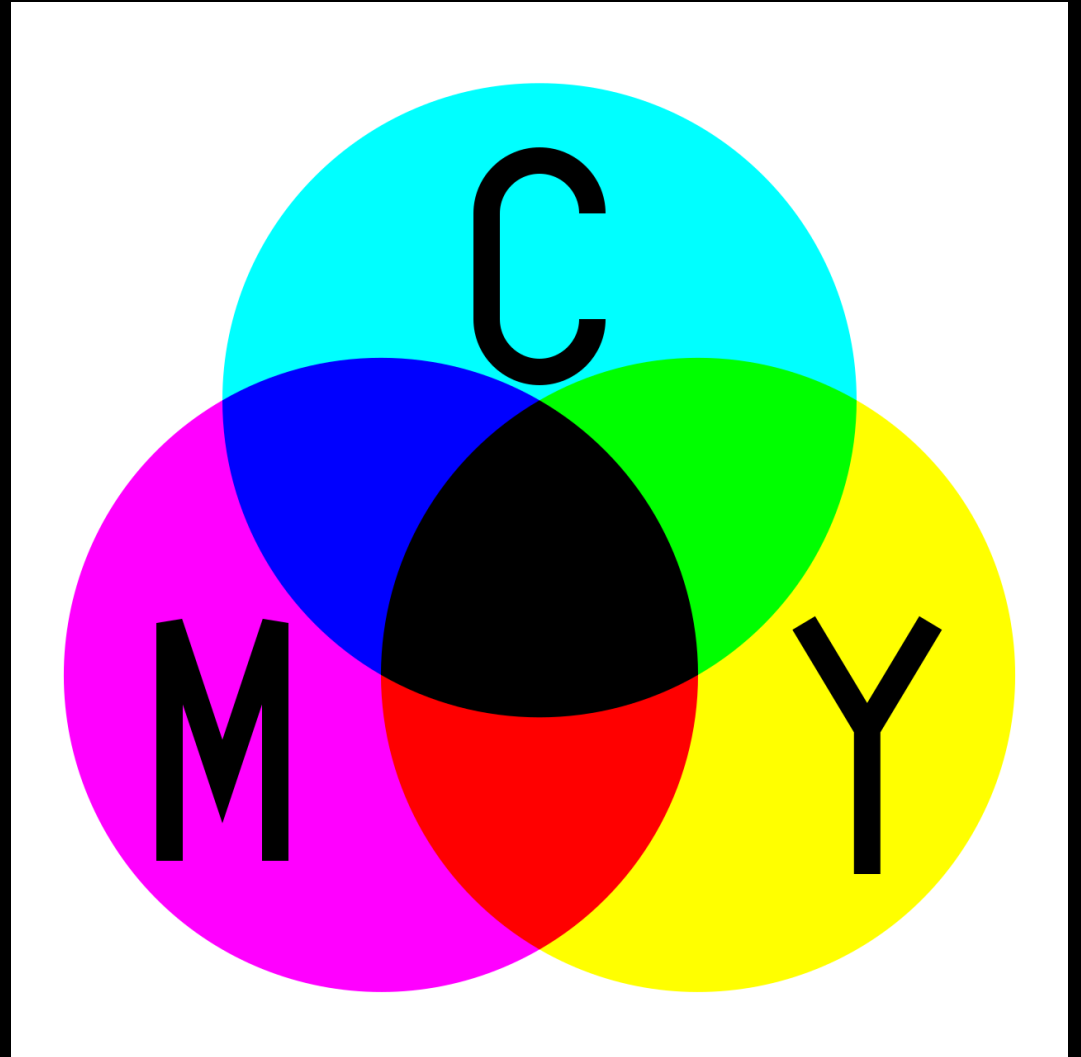
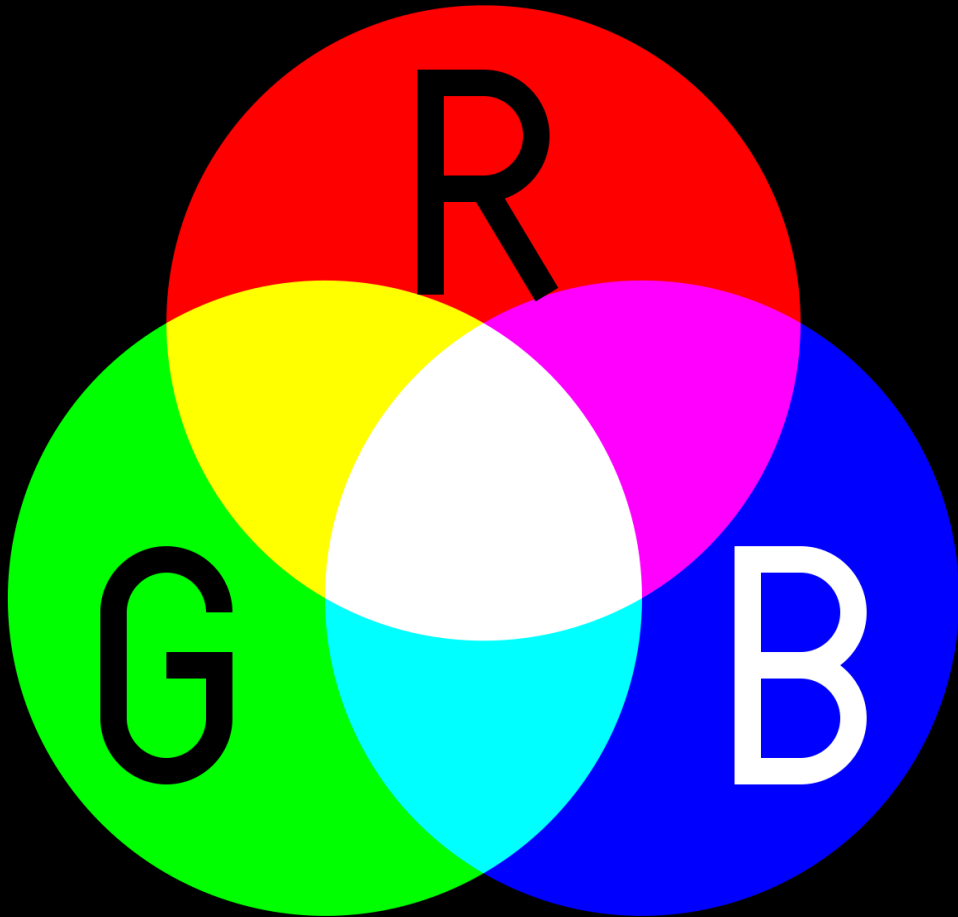
An insertion between 10 and 11:

chr1:10-11 -> chr1 10 10

Color Theory



Additive vs Subtractive Color



RGB Color Model

- Red
- Green
- Blue

RGB Color Model

- Red
 - Green
 - Blue
-
- Red, Green, Blue

RGB Color Model

- Red
 - Green
 - Blue
-
- Red, Green, Blue
 - 0-255, 0-255, 0-255

RGB Color Model

- Red
 - Green
 - Blue
-
- Red, Green, Blue
 - 0-255, 0-255, 0-255
 - 00-FF, 00-FF, 00-FF

RGB Color Model

- Red
 - Green
 - Blue
-
- Red, Green, Blue, Transparency (Alpha Value)
 - 0-255, 0-255, 0-255, 0-255
 - 00-FF, 00-FF, 00-FF, 00-FF

Resources for Choosing Colors

- <https://htmlcolors.com/>
 - Doesn't address accessibility
- <https://colorbrewer2.org/>

BED Format

- This is not PLINK's binary pedigree format
- Browser Extensible Data

BED Format

- 3–12 fields

chrom

Required

chromStart

Required

chromEnd

Required

name

These columns are optional, but “binding”:

score

If you include a column, all rows must include it, and

strand

if you include a column, you must include the

thickStart

columns before it. (Missing data = .)

thickEnd

itemRgb

blockCount

blockSizes

blockStarts

Required Fields (BED3)

- chrom Name of the chromosome (chr1, chr2, chrX, etc.)
- chromStart Start of the feature (0-start)
- chromEnd End of the feature (0-start)
- **Space- or tab-delimited**
 - Tab-delimited is preferable if you are uploading a file
 - Space-delimited is useful if you are typing in data

Metadata Lines

- Browser line(s)
- Track line

Browser Line(s)

- Configure(s) the Genome Browser display
 - Positions shown
 - Tracks shown
 - "Density" settings for tracks

Browser Line(s)

```
browser position chr22:1-10000    # 1-start  
browser hide all  
browser pack knownGene
```

- Separate lines
- Visibility modes: hide, dense, pack, squish, full

Track Line

- Configure the track display
 - Name
 - Type
 - Visibility mode
 - Color
 - etc.

Track Line

```
track name=Example1 description="HUGEN 2073" color=255,0,0  
visibility=3
```

- All one line
- **Visibility modes:** hide=0, dense=1, full=2, pack=3, squish=4

BED3 Example

```
browser position chr22:1-10000  
browser hide all  
track name=Example1 description="HUGEN 2073"  
    color=255,0,0 visibility=3  Make sure track is all one line  
chr22 1000 5000  
chr22 2000 6000
```

Go to **My Data > Custom Tracks** in the UCSC Genome Browser and try adding this track

BED Format

- 3–12 fields

chrom

Required

chromStart

Required

chromEnd

Required

name

These columns are optional, but “binding”

score

If you include a column, all rows must include it, and

strand

if you include a column, you must include the columns

thickStart

before it. (Missing data = .)

thickEnd

itemRgb

blockCount

blockSizes

blockStarts

Optional Fields (BED4 to BED12)

- `name` A label for the feature
- `score` Quantitative value from 0–1000 that will display the feature in shades of gray if `useScore=1`
- `strand` . or + or –
- `thickStart` The starting point for going from a thin line to a thick bar (think TSS), set to `chromStart` for non-genes
- `thickEnd` The starting point for going from a thin line to a thick bar (think TES), set to `chromEnd` for non-genes

Optional Fields (BED4 to BED12)

- `itemRgb` · RGB value 255, 0, 0 that will display the feature in color if `itemRgb=1`
- `blockCount` · Number of thick bars blocks (think exons)
- `blockSizes` · List of block sizes (lengths) exon lengths
- `blockStarts` · List of starting points for blocks, relative to `chromStart`

BED12 Example

```
track name="twoGenes" description="Example Genes"  
  itemRgb=1 visibility=3  Make sure track is all one line  
chr22 1000 5000 geneA 0 + 1000 5000 255,0,0 2 567,488 0,3512  
chr22 2000 6000 geneB 0 - 2000 6000 0,0,255 2 433,399 0,3601
```

bedGraph Format

- Adds a track of continuous values (with colors and shades etc.)

bedGraph Format

- 4 fields

chrom	Required
chromStart	Required
chromEnd	Required
dataValue	Optional

Track Line

```
track type=bedGraph name=Example2 description="HUGEN 2073  
2" color=255,0,0 altColor=0,0,255 visibility=3  
graphType=points
```

- All one line
- Required
- Visibility modes: hide=0, dense=1, full=2, pack=3, squish=4
- Graph types: bar, points

bedGraph Example

```
track type=bedGraph name=Example2
      description="HUGEN 2073 2" color=255,0,0
      altColor=0,0,255 visibility=3
      graphType=points Make sure track is all one line
chr22 1010 1011 10.1
chr22 1011 1025 3
chr22 1025 1030 -2
chr22 1030 1033 -6
chr22 1033 1040 -3
```

Personal Genome SNP Format

- 7 fields

chrom

chromStart

chromEnd

alleleNames

alleleCount

alleleFreq

alleleScores

Personal Genome SNP Format Example

```
track type=pgSnp visibility=3 db=hg38 name="pgSnp"
description="Personal Genome SNP example"
browser position chr22:1924-2937
chr22 2007 2008 T/G 2 21,70 0,0
chr22 2035 2035 -/CGG 2 20,80 0,0
chr22 2088 2093 CTCGG/- 2 30,70 0,0
chr22 2277 2278 T 1 15 0
chr22 2771 2772 A 1 36 0
chr22 2827 2828 A/T 2 15,5 0,0
chr22 2879 2880 C 1 0 0
chr22 2915 2916 - 1 0 0
```